BIO

Carmel Pak U Secondary School
2nd Mid-term Examination 24-25
F.3 Integrated Science

Date of exam: 31/3/2025
Time allowed: 20 minutes

F. 3 ( )
Name
( )

1 A B C D
2 A B C D
3 A B C D
4 A B C D
5 A B C D

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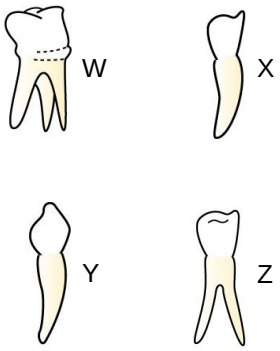
5 question sheet (5169)

/25

A. MC Questions (5 marks)

1. The diagram below shows four types of human teeth.



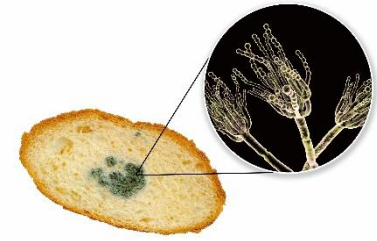
Which type of teeth is ***absent*** in the milk teeth?

A. W
B. X
C. Y
D. Z

2. Which of the following events ensure that the food bolus enters the oesophagus during the process of swallowing?
(1) The soft palate moves up to prevent the bolus from entering the nasal cavity.
(2) The larynx rises so that the epiglottis covers the opening to the trachea.
(3) The tongue rises to push the bolus towards the pharynx.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

3. Which of the following combinations correctly matches the organisms with their modes of nutrition?



	Bread mould	**Plant**	**Dog**
A.	saprophytic	autotrophic	holozoic
B.	saprophytic	holozoic	autotrophic
C.	holozoic	autotrophic	saprophytic
D.	holozoic	saprophytic	autotrophic

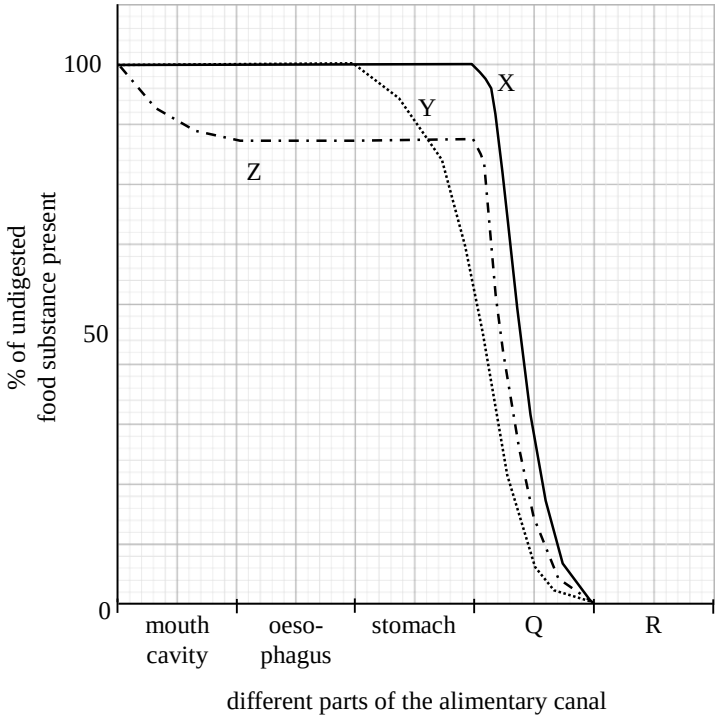
4. Which of the following are the functions of hydrochloric acid in the gastric juice?
   (1) killing bacteria in food
   (2) hydrolysing starch into maltose
   (3) providing an acidic medium for pepsin to work

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3)

5. Which of the following is the function of the cardiac sphincter of the stomach?
   A. To secrete gastric juice.
   B. To shape food into a bolus.
   C. To prevent food from returning to oesophagus.
   D. To control the exit of food from the stomach.

B. Structural question (20 marks)

1. The graph below shows the digestion of three types of food substances (X, Y and Z) along the alimentary canal.



(a) Arrange the pH values of digestive juices found in mouth cavity, stomach and Q in ***descending order***. (1 mark)

_____ > _____ > _____

(b) Name structure R._____(1 mark)

(c) Which curve represents the digestion of protein? Explain your answer. (3 marks)

_____

_____

_____

_____

(d) State and explain how digestive juice(s) is/are responsible for the digestion of lipid in region Q. (4 marks)

_____

_____

_____

_____

_____

(e) Describe how the food bolus passes through the oesophagus. (2 marks)

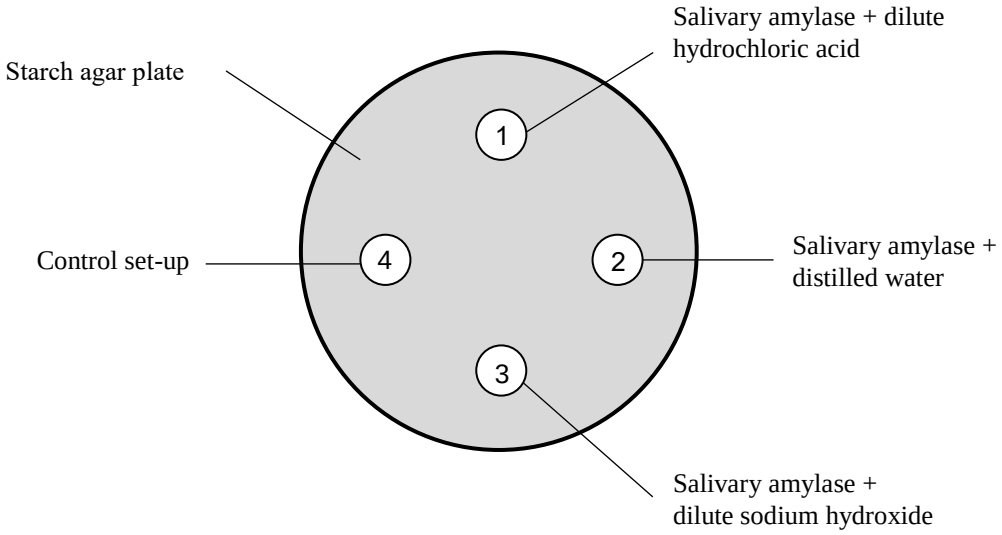
_____

_____

_____

_____

2. In an experiment, four wells are made in a starch agar plate and each well is filled with a different solution as shown in the diagram below.



(a) Identify the variables in the investigation.

(i) Independent variable: _____ (1 mark)

(ii) ***One*** controlled variable: _____ (1 mark)

(b) Explain how to measure the dependent variable. (3 marks)

_____

_____

_____

(c) What should be added to Well 4 to show that the result of Well 1 is caused by the presence of amylase? (2 marks)

_____

_____

(d) Explain which well will the salivary amylase work the best in. (2 marks)

_____

_____

End of paper

MC

DBABC

2. Which of the following events ensure that the food bolus enters the oesophagus during the process of swallowing?
   (1) The soft palate moves up to prevent the bolus from entering the nasal cavity.
   (2) The larynx rises so that the epiglottis covers the opening to the trachea.
   (3) The tongue rises to push the bolus towards the pharynx. → ***this is a process of swallowing but not ensuring the food bolus enter the oesophagus.***
   A. (1) only
   B. (1) and (2) only (15%)
   C. (2) and (3) only
   D. (1), (2) and (3) (65%)

Structural question

1.

(a) Q (small intestine) > mouth cavity > stomach 1m

'Descending order' means from higher pH to lower pH. Acid should have a lower pH while alkaline is higher.

(b) large intestine / colon / rectum 1m

Digestion is completed in small intestine, so the undigested % dropped to ZERO when the food arrived at large intestine.

(c) Y 1m
Protein digestion **starts** in stomach. 1m
Curve Y **starts to drop** in stomach. 1m

Ss cannot simply mention there is digestion of protein as the answer as protein also digested in small intestine.

(d) Bile 1m
contains bile salt for **emulsifying lipid into small droplets** (increase SA). 1m
Pancreatic juice 1m
contains **lipase** to breakdown lipid into **fatty acid and glycerol**. 1m

(e) The two **muscle layers** in the wall of oesophagus **contract alternatively** 1m
to produce a **wave-like movement (peristalsis)** to push the food down. 1m

Total: 11m

2.

(ai) pH value 1m

(ii) amount/volume/concentration of salivary amylase / temperature /
total volume of solution in well / well size / time for digestion 1m

(b) Add iodine solution to the starch agar. 1m
Brown clear zone is formed when starch is digested. 1m
The greater the size / diameter of brown clear zone, the higher the amylase action. 1m

Most Ss did not state how to understand the result by relating the size of clear to the enzyme action.
Ss may mix up this brown clear zone with the clear zone in the milk agar plate.

(c) Prepare an **identical setup/well to well 1** 1m
except **replacing salivary amylase with the same volume of distilled water**. 1m

Ss may mix up the control for 'presence of enzyme' with 'catalytic action of enzyme' and state 'the use of boiled enzyme' as the answer.

(d) Well 2 1m
because the **pH of distilled water** **is similar to** saliva in **mouth cavity**. 1m

Total: 9m